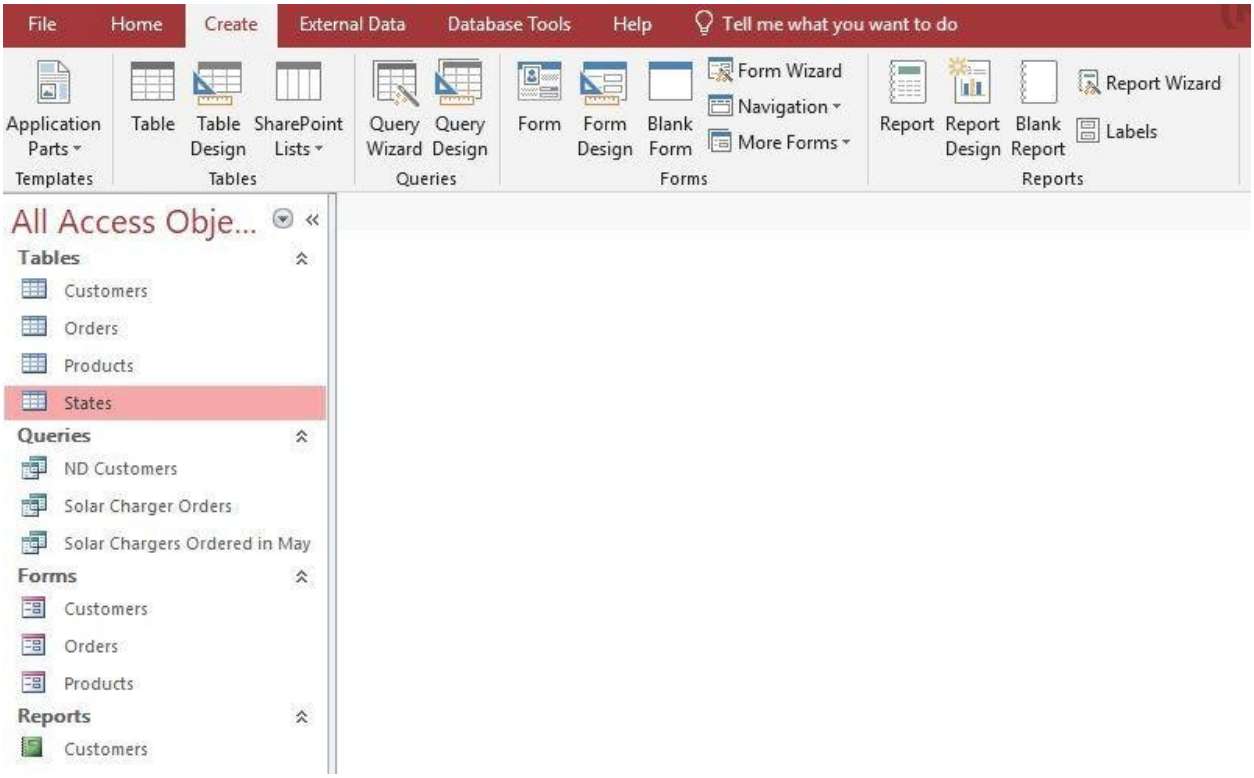
Practical 1: Introduction to MS Access and Database Creation

Aim: To become familiar with MS Access, create a database, design tables, insert records and apply integrity constraints.

A. Getting familiar with MS Access Ribbon options



The MS Access Ribbon is a toolbar at the top of the application that contains commands used to create, manage and manipulate databases.

Ribbon tab	Uses
File	Create, open, save and print databases
Home	Clipboard, sorting, filtering and formatting
Create	Create tables, queries, forms and reports
External Data	Import and export data

Database Tools	Relationships, macros and database analysis
Help	Access help and support

Procedure:

1. Open Microsoft Access.
2. Select Blank Database.
3. Enter a database name.
4. Click Create.
5. Explore the Ribbon tabs and their available commands.

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B. Create a database using the Access Wizard and add two tables

Aim: To create a database with two tables, each containing five fields of different data types and ten records.

Procedure:

1. Open MS Access and create a blank database named CollegeDB.
2. Select Create → Table Wizard (where available) to create the first table.
3. Create a table named Student.
4. Add the following fields.

Table 1: Student

Field name	Data type	Description
StudentID	AutoNumber	Primary key
StudentName	Short Text	Student's name
DOB	Date/Time	Date of birth
Fees	Currency	Student fees
IsActive	Yes/No	Active status

1. Set StudentID as the primary key.
2. Save the table as Student.

Table 2: Course

Field name	Data type	Description
CourseID	AutoNumber	Primary key
CourseName	Short Text	Course name
Duration	Number	Duration in years
StartDate	Date/Time	Course start date
CourseFee	Currency	Course fees

1. Set CourseID as the primary key.

2. Save the table as Course.
3. Enter ten records in each table using Datasheet View.
4. Establish a relationship between the tables if the database design includes a course reference in the student table. For a proper one-to-many relationship, add CourseID as a foreign key in Student.

Integrity constraints:

Primary key: Uniquely identifies each record.

Foreign key: Links related records in different tables.

Required: Prevents a field from being left blank.

Validation rule: Restricts invalid values.

Referential integrity: Prevents orphan records in related tables

Use Table Wizard, add and delete fields, input masks and validation rules

Procedure:

1. Open the database and select Create → Table Wizard.
2. Choose a sample table category and select required fields.
3. Finish the wizard and save the table.
4. Open the table in Design View.
5. Add a new field by entering a field name and data type.
6. Delete an existing field by selecting its row and clicking Delete Rows.
7. Set an input mask and validation rule in the Field Properties section.
8. Switch between Design View and Datasheet View using the View button.

Examples of field properties:

Field	Input mask / validation
Phone	00000-00000
Age	>=18 And <=60

esult: A database with two tables is created, and field properties and table views are successfully managed.

Practical 2: Input Masks, Records and Employee Database

Aim: To use input masks, insert records through different methods and perform update, delete and replace operations.

A. Create and use an Input Mask

An input mask controls the format in which data is entered into a field.

Examples:

Data	Input mask
Phone number	00000-00000
Date	00/00/0000
PIN code	000000
Employee code	LL-0000

Procedure:

1. Open a table in Design View.
2. Select the required field.
3. In Field Properties, select Input Mask.
4. Click the Build button if available.
5. Select or define the required mask.
6. Save the table and enter records in Datasheet View.

B. Adding records using Datasheet View, Form and SQL

1. Using Datasheet View

1. Open the table.

2. Click the new record row marked with an asterisk.
3. Enter values in the fields.
4. Move to another row to save the record.

2. Using a Form

1. Select the table.
2. Go to Create → Form.
3. Enter data into the form fields.
4. Navigate to a new record and enter another record.

3. Using SQL

INSERT INTO Employee

(EmployeeID, EmployeeName, Job, Salary)

VALUES

(101, 'Rahul', 'Manager', 45000);

Run the query in SQL View. The field names and data types must match the actual table structure.

C. Create Employee Database and perform operations

Create a table named Employee.

EmployeeID	EmployeeName	Job	Salary
101	Kelly Marder	Manager	55000
102	Pamela Milgrom	Manager	58000
103	John Smith	Clerk	25000
104	Amit Patil	Manager	48000

105	Sneha Shah	Assistant	30000

1. Delete the record for Kelly Marder

DELETE FROM Employee

WHERE EmployeeName = 'Kelly Marder';

2. Change Pamela Milgrom's salary to $59,500

UPDATE Employee

SET Salary = 59500

WHERE EmployeeName = 'Pamela Milgrom';

3. Replace all occurrences of Manager with Supervisor

UPDATE Employee

SET Job = 'Supervisor'

WHERE Job = 'Manager';

Alternatively, use Home → Find → Replace to replace Manager with Supervisor in the Job field.

Result: Records are inserted, deleted, updated and replaced successfully.

Practical 3: Bookstore Database, SQL Queries and Form Wizard

Aim: To create a Bookstore database, modify records, execute SQL queries and create a form using Form Wizard.

A. Create Bookstore database and modify records

Procedure:

1. Open MS Access and create a blank database named Bookstore.
2. Create a table named Books.
3. Add the following fields.

Field name	Data type

ISBN	Short Text (Primary Key)
BookTitle	Short Text
Author	Short Text
Publisher	Short Text
PublishYear	Number
Price	Currency

4. Enter sample book records.

Operations:

i. Add a new book

Book details:

Title: Exploring Microsoft Office 2000 Vol II

ISBN: 013-011100-7

Author: Grauer/Barber

Publisher: Prentice Hall

Year: 1999

Price: $45.00

```
INSERT INTO Books
(ISBN, BookTitle, Author, Publisher, PublishYear, Price)
VALUES
('013-011100-7',
'Exploring Microsoft Office 2000 Vol II',
'Grauer/Barber', 'Prentice Hall', 1999, 45.00);
```

ii. Change the price of Memory Management for All of Us

```
UPDATE Books
```

```
SET Price = 29.95
WHERE BookTitle = 'Memory Management for All of Us';
```

iii. Delete The Presentation Design Book

```
DELETE FROM Books
WHERE BookTitle = 'The Presentation Design Book';
```

B. Create employ table and execute queries

Create a table named employ.

Field name	Data type
idno	Number (Primary Key)
name	Short Text
job	Short Text
age	Number
salary	Currency

Insert 10 records

```
INSERT INTO employ
(idno, name, job, age, salary)
VALUES
(1, 'Amit', 'Manager', 35, 25000);
```

Repeat the INSERT statement with different values for the remaining nine employees.

Query 1: Display information of all managers

```
SELECT *
FROM employ
WHERE job = 'Manager';
```

Query 2: Display names of employees whose salary is greater than 15000

```
SELECT name
FROM employ
WHERE salary > 15000;
```

C. Create a form using Form Wizard

Procedure:

1. Open the database and select the required table.
2. Click Create → Form Wizard.
3. Select the required fields and move them to Selected Fields.
4. Click Next and select the layout, such as Columnar or Tabular.
5. Choose a suitable style.
6. Enter the form name and click Finish.
7. Open the form in Layout View or Design View.
8. Move controls by dragging them and resize them using their handles.
9. Open the completed form and enter data into the associated table.

Result: The Bookstore database is modified, SQL queries are executed and a form is created for data entry.

Practical 4: Lookup Wizard, Forms, Queries and Expression Builder

Aim: To add fields, create combo boxes, modify forms, generate queries and use Expression Builder.

A. Add fields, Lookup Wizard and command buttons

1. Add fields to an existing table

1. Open the table in Design View.
2. Add a new field name.
3. Select its data type.
4. Save the table.

2. Use Lookup Wizard to create a combo box

A Lookup Wizard creates a drop-down list to select predefined values.

Example: Add a Department field to the Employee table.

Procedure:

1. Open the Employee table in Design View.
2. Add a field named Department.
3. Select Lookup Wizard as the data type.
4. Select the option to enter values manually.
5. Enter values such as IT, HR, Finance and Sales.
6. Finish the wizard and save the table.
7. Open Datasheet View and select a value from the combo box.

3. Add controls to an existing form to demonstrate inheritance

Controls in a form can be duplicated or reused to maintain consistent formatting and structure.

1. Open an existing form in Design View.
2. Select an existing control, such as a text box.
3. Copy and paste it or use the appropriate control from the Controls group.
4. Set its Control Source to the required field.
5. Modify its properties while retaining a consistent layout and appearance.

4. Add command buttons to a form

1. Open the form in Design View.
2. Select Design → Button.
3. Click the desired location on the form.
4. Select an action in the Command Button Wizard, such as Add New Record, Save Record, Delete Record or Close Form.
5. Enter a caption and finish the wizard.
6. Save and test the form.

B. Generate queries using Query Wizards

A query is used to retrieve, filter, calculate and analyze data from tables.

Procedure:

1. Open the database.
2. Select Create → Query Wizard.
3. Select Simple Query Wizard.
4. Choose the table and fields.
5. Select Detail or Summary as required.
6. Enter a query name and click Finish.

Example: Display employees with salary greater than 20000

```
SELECT *
FROM Employee
WHERE Salary > 20000;
```

Other queries can be created using the Crosstab Query Wizard and Find Duplicates Query Wizard, depending on the task.

C. Generate queries with user input

A parameter query allows the user to enter a value when the query is executed.

Example: Display employees by job

```
SELECT *
FROM Employee
WHERE Job = [Enter Job Title];
```

Procedure:

1. Select Create → Query Design.
2. Add the Employee table.
3. Switch to SQL View.
4. Enter the parameter query.
5. Save and run the query.
6. Enter a job title when prompted.

The query displays only employees matching the entered job title.

D. Demonstrate the use of Expression Builder

Expression Builder helps create calculated expressions for queries, forms and reports.

Example 1: Calculate annual salary

AnnualSalary: [Salary]*12

Example 2: Calculate employee age category

AgeCategory: IIf([Age]>=18,"Adult","Minor")

Procedure:

1. Open a query in Design View.
2. Select an empty Field row.
3. Right-click and select Build, or click Expression Builder from the Ribbon.
4. Enter the required expression.
5. Click OK and save the query.
6. Run the query to view the calculated results.

Result: Lookup fields, form controls, parameter queries and calculated expressions are created successfully.

Practical 5: Reports, Calculated Controls, Action Queries and Crosstab Queries

Aim: To create and modify reports, generate calculated fields, group report data, use action queries and create crosstab queries.

A. Create and modify a report using Report Wizard

A report is used to present database information in a structured, printable format.

Procedure:

1. Open the database.
2. Select the required table or query.
3. Click Create → Report Wizard.
4. Select the fields to include in the report.
5. Select grouping levels, if required.
6. Choose a sorting order.

7. Select a layout and orientation.
8. Enter the report name and click Finish.
9. Open the report in Design View to modify it.

Modifying an existing report:

Add controls such as labels, text boxes and images.

Delete unnecessary controls.

Change the size, position, font and color of controls.

Add a title and date.

Save and preview the report.

B. Create a query with a calculated control and generate a report

Example: Calculate the total salary of employees, including a bonus of 10%.

SQL Query:

```
SELECT EmployeeID,
    EmployeeName,
    Salary,
    Salary*0.10 AS Bonus,
    Salary*1.10 AS TotalSalary
FROM Employee;
```

Procedure:

1. Create a query in Query Design.
2. Add the Employee table.
3. Select the required fields.
4. Add the calculated fields.
5. Save the query as SalaryCalculation.
6. Run the query to verify the results.
7. Select the query and click Create → Report Wizard.
8. Select the required fields and complete the wizard.

Sorting and Grouping:

1. Open the report in Design View.
2. Select Design → Group & Sort.
3. Add a grouping field, such as Department.
4. Enable the group header and group footer.
5. Add a label to the header and a calculated summary to the footer, such as a sum of salaries.
6. Save and preview the report.

C. Use action queries to modify a database

Action queries perform operations such as updating, deleting, appending and creating tables.

Action query	Purpose
Update	Modifies existing records
Delete	Deletes selected records
Append	Adds records to another table
Make-Table	Creates a new table from query results

1. Update Query

UPDATE Employee

SET Salary = 50000

WHERE EmployeeID = 101;

2. Delete Query

DELETE FROM Employee

WHERE EmployeeID = 105;

3. Append Query

INSERT INTO EmployeeArchive

(EmployeeID, EmployeeName, Salary)

SELECT EmployeeID, EmployeeName, Salary

FROM Employee

WHERE Salary > 40000;

4. Make-Table Query

SELECT *

INTO HighSalaryEmployees

FROM Employee

WHERE Salary > 40000;

Run action queries carefully, as changes to records may not be easily reversible. Always back up important data before executing them.

D. Create a Crosstab Query

A crosstab query summarizes data in a row-and-column format, making comparisons easier.

Example: Summarize employee salaries by department and job.

TRANSFORM Sum(Salary) AS TotalSalary

SELECT Department

FROM Employee

GROUP BY Department

PIVOT Job;

Procedure:

1. Select Create → Query Wizard.
2. Choose Crosstab Query Wizard.
3. Select the table or query.
4. Select the row heading, such as Department.
5. Select the column heading, such as Job.
6. Select the value field, such as Salary, and choose Sum.

7. Enter a query name and finish.
8. View the summarized data.

Note: The SQL example assumes the Employee table contains Department and Job fields.

Result: Reports are created and modified, calculated controls are displayed, action queries are executed and summarized data is generated using a crosstab query.